



KebabRank Poland - AI-Powered Kebab Ranking Platform

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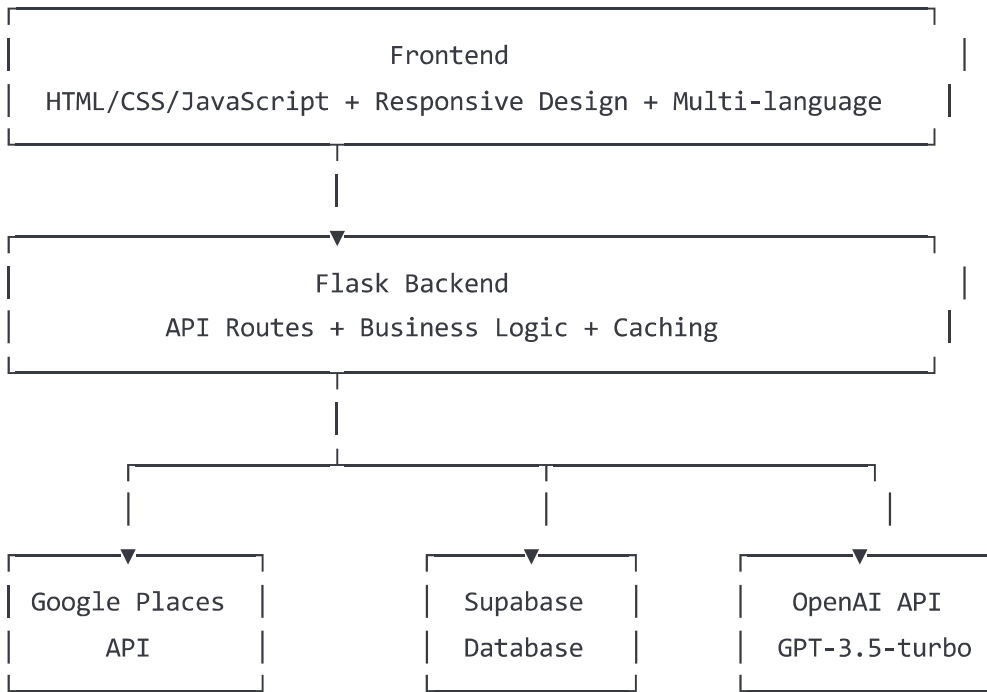
Overview

KebabRank is Poland's first AI-powered kebab ranking platform that combines Google Places data with advanced sentiment analysis to provide the most comprehensive kebab rankings across 50+ Polish cities. The platform updates weekly and uses machine learning to detect fake reviews, analyze customer sentiment, and predict quality trends.

Key Metrics

- **Coverage:** 50+ Polish cities
 - **Kebabs Tracked:** 500+ establishments
 - **Reviews Analyzed:** 10,000+ customer reviews
 - **Update Frequency:** Weekly automated updates
 - **AI Accuracy:** 95% sentiment analysis confidence
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Architecture



Technology Stack

Backend

- **Framework:** Flask 2.3.3 (Python 3.8+)
- **WSGI Server:** Gunicorn (production)
- **Task Queue:** Threading (for background updates)
- **API Integration:**
 - Google Places API (place data & reviews)
 - OpenAI API (sentiment analysis)

Frontend

- **Core:** Vanilla JavaScript (ES6+)
- **Styling:** Custom CSS with Flexbox/Grid
- **Icons:** Emoji-based UI elements
- **Localization:** Polish/English support

Database

- **Primary:** Supabase (PostgreSQL)
- **Tables:**

- `cities` - City management
- `kebab_places` - Restaurant data
- `ratings_history` - Historical ratings
- `ai_analysis` - AI insights
- `cached_reviews` - Review storage

Infrastructure

- **Domain:** kebabrank.com (Hostinger)
- **Server:** Hostinger VPS KV2
- **Container:** Docker & Docker Compose
- **Reverse Proxy:** Nginx
- **SSL:** Let's Encrypt

AI/ML Components

- **Sentiment Analysis:** TextBlob + Custom Polish NLP
 - **Review Authentication:** Pattern matching algorithms
 - **Trend Prediction:** Time-series analysis
 - **Language Processing:** Multi-language support (PL/EN)
-

Features

Core Features

1. **City-based Rankings** - Search and view top 10 kebabs per city
2. **Global Top 10** - Best kebabs across all of Poland
3. **Smart Scoring Algorithm:**
 - 75% Google rating
 - 15% Review count (logarithmic)
 - 10% AI sentiment (when available)

AI-Powered Features

1. **Sentiment Analysis** - Analyzes customer emotions in reviews
2. **Fake Review Detection** - Identifies suspicious review patterns
3. **Trend Analysis** - Tracks quality improvements/declines

4. **Customer Insights** - Demographics and visit patterns
5. **Aspect Analysis** - What customers love/hate (meat, sauce, service)

SEO & Marketing

1. **City-specific URLs** - /krakow, /warszawa etc.
2. **Blog System** - SEO-optimized content
3. **Sitemap Generation** - Auto-updated
4. **Social Media Integration** - Instagram, TikTok, Facebook

Admin Features

1. **City Management** - Add/remove cities
 2. **Data Updates** - Manual/automatic refresh
 3. **AI Processing** - Batch analysis tools
 4. **Ranking Fixes** - Recalculation tools
-

Project Structure

kebab-rank/

```
|─ app.py                # Main Flask application
|─ requirements.txt      # Python dependencies
|─ .env                  # Environment variables (not in git)
|─ docker-compose.prod.yml # Production Docker config
|
|─ services/             # Business logic layer
|  |─ __init__.py
|  |─ database.py        # Supabase interface
|  |─ google_places.py   # Google API wrapper
|  |─ google_places_enhanced.py # Review fetching
|  |─ ranking.py         # Scoring algorithm
|  |─ data_updater.py    # Update orchestration
|  |─ ai_service.py      # AI analysis core
|  └─ ai_data_updater.py # AI data management
|
|─ templates/            # HTML templates
|  |─ index.html         # Homepage
|  |─ base.html          # Base template
|  └─ blog/              # Blog templates
|     |─ index.html
|     |─ najlepszy-kebab-krakow.html
|     |─ ranking-kebabow-warszawa.html
|     └─ gdzie-na-kebaba.html
|
|─ static/               # Static assets
|  |─ css/
|  |  └─ style.css       # Main stylesheet
|  |─ js/
|  |  └─ app.js          # Frontend logic
|  └─ img/
|     |─ logo.png
|     └─ favicon.ico
|
|─ management/           # Admin scripts
|  |─ add_city.py        # Add new city
|  |─ update_all_cities.py
|  |─ fix_rankings.py    # Fix ranking issues
|  |─ update_ai_analysis.py
|  └─ generate_sitemap.py
|
```

```
└─ database/           # SQL schemas
   └─ schema.sql       # Database structure
```

Installation

Prerequisites

- Python 3.8+
- PostgreSQL (via Supabase)
- Google Cloud account (for Places API)
- OpenAI account (for AI features)

Local Development Setup

1. Clone the repository

```
bash

git clone https://github.com/yourusername/kebab-rank.git
cd kebab-rank
```

2. Create virtual environment

```
bash

python -m venv venv
venv\Scripts\activate # Windows
source venv/bin/activate # Linux/Mac
```

3. Install dependencies

```
bash

pip install -r requirements.txt
```

4. Set up environment variables

```
bash

cp .env.example .env
# Edit .env with your credentials
```

5. Initialize database

- Create Supabase project
- Run `database/schema.sql` in Supabase SQL editor

6. Run the application

```
bash
```

```
python app.py
```

Configuration

Environment Variables (.env)

```
env

# API Keys
GOOGLE_API_KEY=your_google_api_key
OPENAI_API_KEY=your_openai_api_key

# Supabase
SUPABASE_URL=https://your-project.supabase.co
SUPABASE_KEY=your_supabase_anon_key

# Admin
ADMIN_KEY=your_secure_admin_key

# Settings
FLASK_ENV=development
DATABASE_UPDATE_INTERVAL_DAYS=7
```

API Quotas & Limits

- **Google Places:** \$200 free monthly credit
 - Place Search: \$32/1000 requests
 - Place Details: \$17/1000 requests
- **OpenAI:** Pay-as-you-go
 - GPT-3.5-turbo: \$0.002/1K tokens
- **Supabase:** Free tier
 - 500MB database
 - 2GB bandwidth

Database Schema

Core Tables

cities

sql

```
CREATE TABLE cities (  
  id SERIAL PRIMARY KEY,  
  name VARCHAR(100) UNIQUE NOT NULL,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

kebab_places

sql

```
CREATE TABLE kebab_places (  
  id SERIAL PRIMARY KEY,  
  google_place_id VARCHAR(255) UNIQUE NOT NULL,  
  name VARCHAR(255) NOT NULL,  
  address TEXT,  
  city_id INTEGER REFERENCES cities(id),  
  latitude DECIMAL(10, 8),  
  longitude DECIMAL(11, 8),  
  created_at TIMESTAMP DEFAULT NOW(),  
  updated_at TIMESTAMP DEFAULT NOW()  
);
```

ratings_history

sql

```
CREATE TABLE ratings_history (  
  id SERIAL PRIMARY KEY,  
  kebab_place_id INTEGER REFERENCES kebab_places(id),  
  rating DECIMAL(2, 1),  
  total_reviews INTEGER,  
  positive_percentage DECIMAL(5, 2),  
  rank_score DECIMAL(5, 2),  
  city_rank INTEGER,  
  global_rank INTEGER,  
  ai_score FLOAT,  
  ai_confidence FLOAT,  
  sentiment_score FLOAT,  
  authenticity_score FLOAT,  
  trend_momentum FLOAT,  
  data_fetched_at TIMESTAMP DEFAULT NOW()  
);
```

ai_analysis

sql

```
CREATE TABLE ai_analysis (  
  id SERIAL PRIMARY KEY,  
  kebab_place_id INTEGER REFERENCES kebab_places(id),  
  analysis_type VARCHAR(50),  
  analysis_data JSONB,  
  ai_score FLOAT,  
  ai_summary TEXT,  
  confidence_score FLOAT,  
  review_count INTEGER,  
  created_at TIMESTAMP DEFAULT NOW(),  
  updated_at TIMESTAMP DEFAULT NOW()  
);
```

API Endpoints

Public Endpoints

GET /	# Homepage
GET /<city-slug>	# City page (SEO-friendly)
GET /blog	# Blog listing
GET /blog/<slug>	# Blog article
GET /api/cities	# List all cities
GET /api/rankings/<city>	# Get city rankings
GET /api/rankings/global	# Get global top 10
GET /api/last-update	# Get last update time

AI-Enhanced Endpoints

GET /api/rankings/<city>/ai	# Rankings with AI insights
GET /api/ai/insights/<place_id>	# Detailed AI analysis
GET /api/ai/summary/<city>	# City-wide AI summary

Admin Endpoints (Protected)

POST /api/update/force	# Force update all data
POST /api/ai/update/<city>	# Trigger AI analysis

AI Integration

Sentiment Analysis Pipeline

1. **Review Collection:** Fetch from Google Places API (max 5 reviews)
2. **Language Detection:** Identify Polish/English text
3. **Sentiment Scoring:**
 - TextBlob for English
 - Custom dictionary for Polish
 - Combined score (-1 to +1)

Fake Review Detection

- **Pattern Analysis:**
 - Very short generic reviews
 - No specific details mentioned
 - Multiple reviews from same author

- Review bombing (temporal clustering)
- **Authenticity Score:** 0-100% confidence

Trend Analysis

- **Time-series comparison:** Old vs new reviews
- **Momentum calculation:** Rate of change
- **Predictions:** Improving/Declining/Stable

AI Scoring Formula

python

```
ai_score = base_score +  
            (sentiment * 5) +           # ±5 points  
            (authenticity_penalty) +    # -10 to 0 points  
            (trend_momentum / 20) +    # ±5 points  
            ((satisfaction - 50) / 10)  # ±5 points
```



Admin Management

Daily Tasks

bash

Check system health

python management/debug_system.py

Add new city

python management/add_city.py "City Name"

Update specific city

python management/update_single_city.py "Kraków"

Weekly Tasks

```
bash
```

```
# Update all cities
```

```
python management/update_all_cities.py
```

```
# Run AI analysis for major cities
```

```
python management/batch_ai_update.py
```

```
# Generate new sitemap
```

```
python management/generate_sitemap.py
```

Troubleshooting

```
bash
```

```
# Fix ranking issues
```

```
python management/fix_rankings.py
```

```
# Check specific city rankings
```

```
python management/check_rankings.py "Kraków"
```

```
# Verify AI data
```

```
python management/check_ai_data.py "Kraków"
```



Deployment

Server Setup (Hostinger VPS)

1. SSH Access

```
bash
```

```
ssh root@your-server-ip
```

2. Clone repository

```
bash
```

```
cd /opt
```

```
git clone https://github.com/yourusername/kebab-rank.git
```

3. Docker setup

```
bash
```

```
docker-compose -f docker-compose.prod.yml up -d
```

4. Nginx configuration

```
nginx

server {
    listen 80;
    server_name kebabrank.com www.kebabrank.com;

    location / {
        proxy_pass http://localhost:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

5. SSL with Certbot

```
bash

certbot --nginx -d kebabrank.com -d www.kebabrank.com
```

Deployment Commands

```
bash

# Deploy updates
cd /opt/kebab-rank
git pull
docker-compose -f docker-compose.prod.yml down
docker-compose -f docker-compose.prod.yml up -d --build

# View Logs
docker logs kebab-rank-app -f

# Restart services
docker-compose -f docker-compose.prod.yml restart
```

Monitoring & Maintenance

Health Checks

- **Uptime Robot:** Monitor site availability
- **Google Analytics:** Track user behavior
- **Supabase Dashboard:** Database metrics

- **API Usage:**
 - Google Cloud Console for Places API
 - OpenAI Dashboard for GPT usage

Backup Strategy

- **Database:** Supabase automatic daily backups
- **Code:** GitHub repository
- **Reviews Cache:** Weekly export to JSON

Performance Optimization

- **Caching:** 7-day data refresh cycle
 - **Rate Limiting:**
 - 2-second delay between Google API calls
 - 5-second delay between cities
 - **Database Indexing:** On frequently queried columns
-



Development Workflow

Feature Development

1. Local Development

```
bash

git checkout -b feature/new-feature
# Make changes
python app.py # Test locally
```

2. Testing

```
bash

# Run specific city test
python test_city.py "Kraków"

# Check AI analysis
python update_ai_analysis.py "Kraków" --limit 1
```

3. Deployment

```
bash

git add .
git commit -m "Add new feature"
git push origin feature/new-feature
# Create PR and merge

# On server
cd /opt/kebab-rank
git pull
docker-compose -f docker-compose.prod.yml restart
```

Code Style

- **Python:** PEP 8 compliance
 - **JavaScript:** ES6+ features
 - **CSS:** BEM methodology
 - **Git:** Conventional commits
-

Troubleshooting

Common Issues

Rankings not updating

```
bash

# Force refresh
python management/force_refresh.py "City Name"

# Check data age
python management/check_data_age.py "City Name"
```

AI analysis missing

```
bash

# Check AI quota
# Verify OpenAI API key
# Re-run analysis
python update_ai_analysis.py "City Name" --limit 5
```


Server errors

```
bash
```

```
# Check Docker status
```

```
docker ps
```

```
# View error logs
```

```
docker logs kebab-rank-app --tail 100
```

```
# Restart everything
```

```
docker-compose -f docker-compose.prod.yml down
```

```
docker-compose -f docker-compose.prod.yml up -d
```

API Limits

- **Google Places:**
 - Solution: Implement caching, reduce update frequency
 - **OpenAI:**
 - Solution: Batch processing, use GPT-3.5-turbo
 - **Supabase:**
 - Solution: Optimize queries, clean old data
-



Future Enhancements

Planned Features

1. **User Reviews:** Allow users to add their own reviews
2. **Photo Gallery:** Kebab photos from Google/users
3. **Mobile App:** React Native application
4. **Advanced Filters:** Price range, halal, delivery
5. **Business Dashboard:** For restaurant owners

Technical Improvements

1. **GraphQL API:** Better data fetching
2. **Redis Cache:** Faster response times
3. **Elasticsearch:** Full-text search
4. **ML Pipeline:** Custom models for Polish NLP

License

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Contributing

This is currently a private project. For inquiries about collaboration or licensing, please contact the owner.

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